Band-1 Hadamard Order 4132 — Miyamoto (Paley-I seed H_1032); Gap 2052 Still OPEN

H_1032/C_1034/H_4132 PASS; 4132 Miyamoto-only bonus (NOT in SPK); gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

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Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-15_Band1_Hadamard_4084_Miyamoto`; `2026-09-15_Band1_Hadamard_3716_Miyamoto` (Miyamoto-only pattern).
Lab. `/workspace/experiments/band1/hadamard_4132_miyamoto/`.
Cutoff. 2026-09-15T14:29:26Z (~10:29AM ET).

Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 1032 and 4132 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with Paley-I seed H_{1032} ($q = 1031$). It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under Sylvester∪Paley∪Kronecker (SPK) closure. Order 4132 is a **Miyamoto-only bonus (NOT in SPK)**: Paley-I **FAIL** ($q = 4131 = 3^5 \cdot 17$ not prime); Paley-II **FAIL** ($q = 2065 = 5 \cdot 7 \cdot 59$ not prime); Kronecker **FAIL** ($4132 = 4 \cdot 1033$, 1033 not Hadamard-admissible). Paley-II path for seed H_{1032} **SKIP** ($q = 515 = 5 \cdot 103$ not prime). `clay_closed = false`.

Abstract

Prior Band-1 ships constructed Miyamoto-eligible orders through 4084 on box. This increment ships **order 4132** ($4132 = 4 \cdot 1033$, 1033 prime $\equiv 1 \pmod 4$) via Miyamoto with a direct **Paley-I seed** H_{1032} ($q = 1031$). Symmetric Paley conference C_{1034} over prime 1033; $H_{4132} = \text{Miyamoto}(H_{1032}, C_{1034})$. Unit tests **30/30 PASS**. SPK next gap remains **2052** (OPEN). Order 4132 is **not** in SPK (Miyamoto-only bonus). The **full Hadamard conjecture remains OPEN**.

Keywords. Hadamard matrix; order 4132; Miyamoto; Paley type I; Paley conference; constructive gap 2052; SPK; Band-1; Miyamoto-only bonus; non-claim.

1. Statement lock

Definition. A Hadamard matrix of order n is an $n \times n$ matrix H with entries in ± 1 satisfying $HH^\top = nI$.

Finite lemmas (this increment). 1. H_{1032} constructed and verified (Paley type I, $q = 1031$). 2. C_{1034} constructed and verified (Paley conference, $q = 1033$): $CC^\top = 1033I$, zero diagonal. 3. H_{4132} constructed and verified (Miyamoto). **Not** the Hadamard conjecture. 4. SPK: $4132 \notin$ SPK (Miyamoto-only). Paley-I/II/Kronecker paths for 4132 **rejected**. Next constructive gap remains **2052** (OPEN).

2. Arithmetic and eligibility

Quantity	Value
Target order	$4132 = 4 \cdot 1033$
$q = 1033$	prime, $1033 \equiv 1 \pmod 4 \rightarrow$ Miyamoto-eligible
Needed H_{q-1}	H_{1032}
Needed conference	order $1034 = q + 1$
H_{1032} path (used)	Paley type I, $q = 1031$ (prime $\equiv 3 \pmod 4$); $q + 1 = 1032$
H_{1032} Paley-II	**SKIP**: $2(q + 1) = 1032 \Rightarrow q = 515 = 5 \cdot 103$ not prime
Conference path	Symmetric Paley over prime 1033
Paley-I for 4132	**REJECTED**: $q' = 4131 = 3^5 \cdot 17$ not prime
Paley-II for 4132	**REJECTED**: $q' = 2065 = 5 \cdot 7 \cdot 59$ not prime
Kronecker for 4132	**REJECTED**: only 4×1033; $1033 \not\equiv 0 \pmod 4$ (not admissible)

3. Construction results

Object	Method	$\max \lvert HH^\top - nI \rvert$ (or conf)	Wall (s)	Status
H_{1032}	Paley type I, $q = 1031$	0	1.061	**PASS**
C_{1034}	Paley conference, $q = 1033$	0	0.9928	**PASS**
H_{4132}	Miyamoto(H_{1032}, C_{1034})	0	30.6001	**PASS**

Primary pipeline wall (incl. unit tests + SPK refresh) ≈ 33.8651 s. Compressed ± 1 matrices saved as `H1032_pm1.npz`, `C1034_pm1.npz`, `H4132_pm1.npz` (int8).

4. SPK closure note

Seeds: all admissible $\leq 2000 \cup$ on-box prior classical/SPK ships (through 4036/3908/3748/… + **1020** Paley-II; **exclude** 4084/3716/3524/3284/3076/3044/2836/2692/2564 Miyamoto-only) $\cup$ verified H_{1032} (Paley-I) $\cup$ Sylvester $\cup$ Paley I/II, closed under Kronecker (limit 4500). **Miyamoto is not folded into SPK**; order 4132 remains a Miyamoto-only bonus.

Quantity	Value
Known count	964
4132 in SPK	**False** (Miyamoto-only)
1032 in SPK	**True** (Paley-I)
Next constructive gap	**2052** (OPEN)
Miyamoto-only bonus?	**Yes** (like 4084/3716/3524/…)

5. Gap 2052 (OPEN; method triage / non-finding ONLY)

Path	Result
Paley-I ($q = 2051$)	FAIL
Paley-II ($q = 1025$)	FAIL
Kronecker	FAIL ($2052 = 2^2 \cdot 3^3 \cdot 19$)
Miyamoto + Paley conference ($q = 513$)	FAIL (513 not prime)
GS / SDS / Williamson / Yamada	lit-only; not constructed this run

Status: OPEN. Non-finding $\neq$ counterexample. Conjecture remains OPEN.

6. Unit tests

30/30 PASS (Sylvester/Paley/Miyamoto controls; $q = 1031/1033$ arithmetic; Paley-I seed for H_{1032}; Paley-II skip; SPK path rejection for 4132; Miyamoto-only arith guard; onbox seeds include 1020 / exclude 4084; reject-all-ones; conference control; gap-2052-not-filled guard).

7. Non-claims (explicit)

- No full Hadamard conjecture claim.
- No Clay-closed stamp; `clay_closed = false`.
- No finished-ToE claim; GATE-04 **LOCKED**; no invented J.
- No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims.
- Gap **2052** remains **OPEN** under SPK.
- Order 4132 is a **Miyamoto-only bonus (NOT in SPK)**.

8. Next optional

method-attack SPK gap **2052** OR next Miyamoto ($q = 1049 \to n = 4196$ Double(PaleyI half=524); or $q = 1093 \to n = 4372$ PaleyI(q=1091)) OR Track B privacy / Object 1 meter await.

Lab: `/workspace/experiments/band1/hadamard_4132_miyamoto/` · *Script:* `verify_hadamard_4132.py` · *Artifacts:* `results.json`, `SUMMARY.txt`, `literature_ledger.json`, `*.npz`.